# Polymeric Chemistry
# DSC and TGA

**Universität Stuttgart**

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Assistentin:

## Abstract

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# 1 Results

Two different samples were prepared, and analyzed, using DSC and TGA. The Data of the samples (PET and it-PS) are listed in the Table below 1

Tab. 1: These are the meltingtemperatures($T_m$), glastemperatures($T_g$) and the meltingenthalpie ($\Delta$ H$_m$)

| Sample | $\Delta$ H$_m$ | T$_m$ | T$_g$ |
|---|---|---|---|
| PE | | | |
| *it*-PS | - | | |

# 2 Questions

## 2.1 Question 1

The "Big"number indicates how many monomers are needed to make one full 360° rotation. The small number in the index says how many 360° rotations are needed until the order gets repeated. For example in the case of $8_5$ Polybutene. It needs 8 monomers to complete one 360° rotation and 5 full rotation until the order starts from the start.

## 2.2 Question 2

i-PP can be made by using the Ziegler-Natta-catalysis. In this case catalysts like $TiCl_3$ on $MgCl_2$-carrier are used in combination with aluminumalkenes. It can also be done via Metallocene-catalysis, this technique is specially used to control the tacticity and results in high graded i-PP. i-PS can also be synthesized using the Metallocene-catalysis. But can also be synthesized via Anionic-Polymerization.

## 2.3 Question 3

Commonly the degree of crystallization decides how stiff a polymer is. The higher the degree of crystallization, the harder and stiffer the polymer. It also gets less ductile and more dense because crystalline parts can be packed better than amorph parts.

## 2.4 Question 4

The Polymer is divided in different sections a back folding section(black), which would be the PS in our case and a amorph section(red), which in our case would be the PE sections.

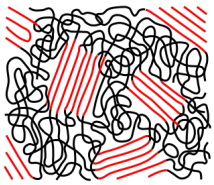


Abb. 1: The picture shows the part-.crystalline

## 2.5 Question 5

To get an semi-crystalline polymer out of an polymerglass it has to be reheated. By reheating the polymer the molecular mobility gets high enough to form crystals. This process is also known as cold crystallization.

## 2.6 Question 6

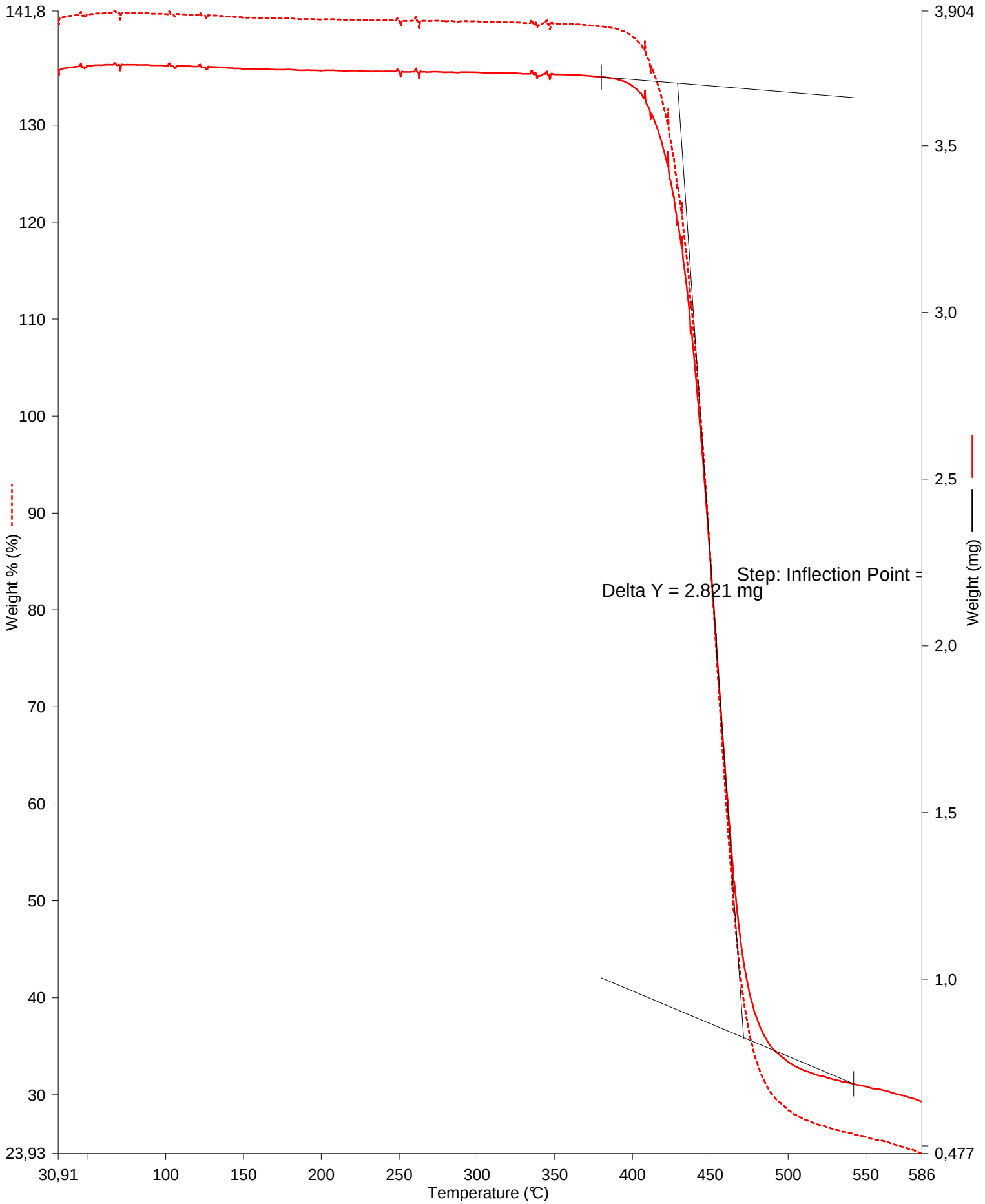
The increase in mass can be explained by oxidation. While increasing the temperature in the TGA the polymer oxidizes, the added Oxygen increases the mass of the polymer (molecule if the polymer gets destroyed).

## 2.7 Question 7

## 2.8 Question 8

Due to the residual moisture, the PET decomposes to Ethylenglycol and Terephthalic acid. The Ethylenglycol eliminates water which results in vinylalcohol, which can tautomerize to Ethanal.

| | |
|---|---|
| Filename: | C:\Users\DSC\Desktop\MD\G...\G7-pet-tga.t6d |
| Operator ID: | MD |
| Sample ID: | G7-PET |
| Sample Weight: | 2.644 mg |
| Comment: | |

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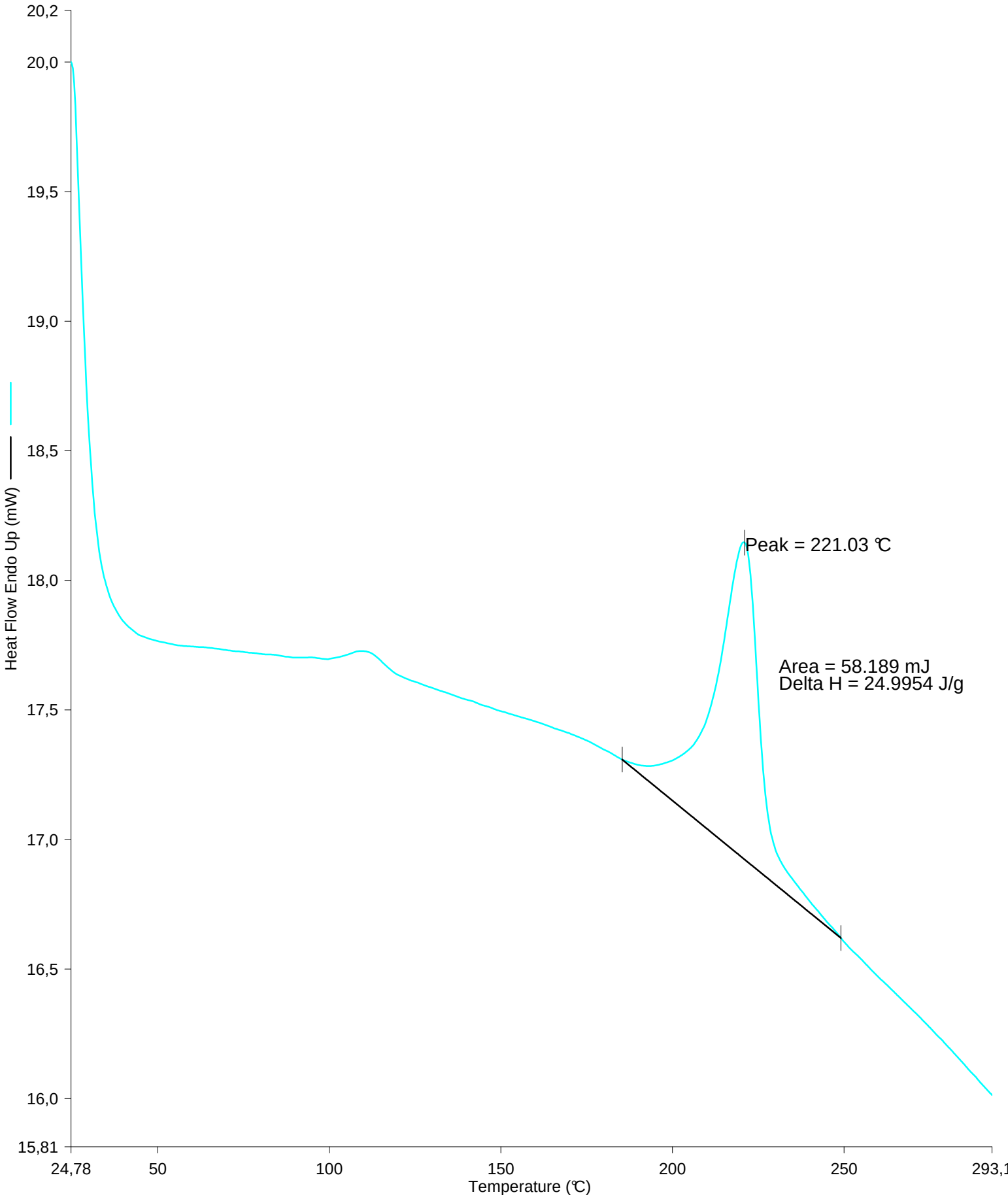
1) Heat from 30.00℃ to 600.00℃ at 20.00℃/min

| | |
|---|---|
| Filename: | C:\Users\DSC\Desktop\MD\G7\isops.d6d |
| Operator ID: | MD |
| Sample ID: | G7-iso-PS |
| Sample Weight: | 2.328 mg |
| Comment: | |

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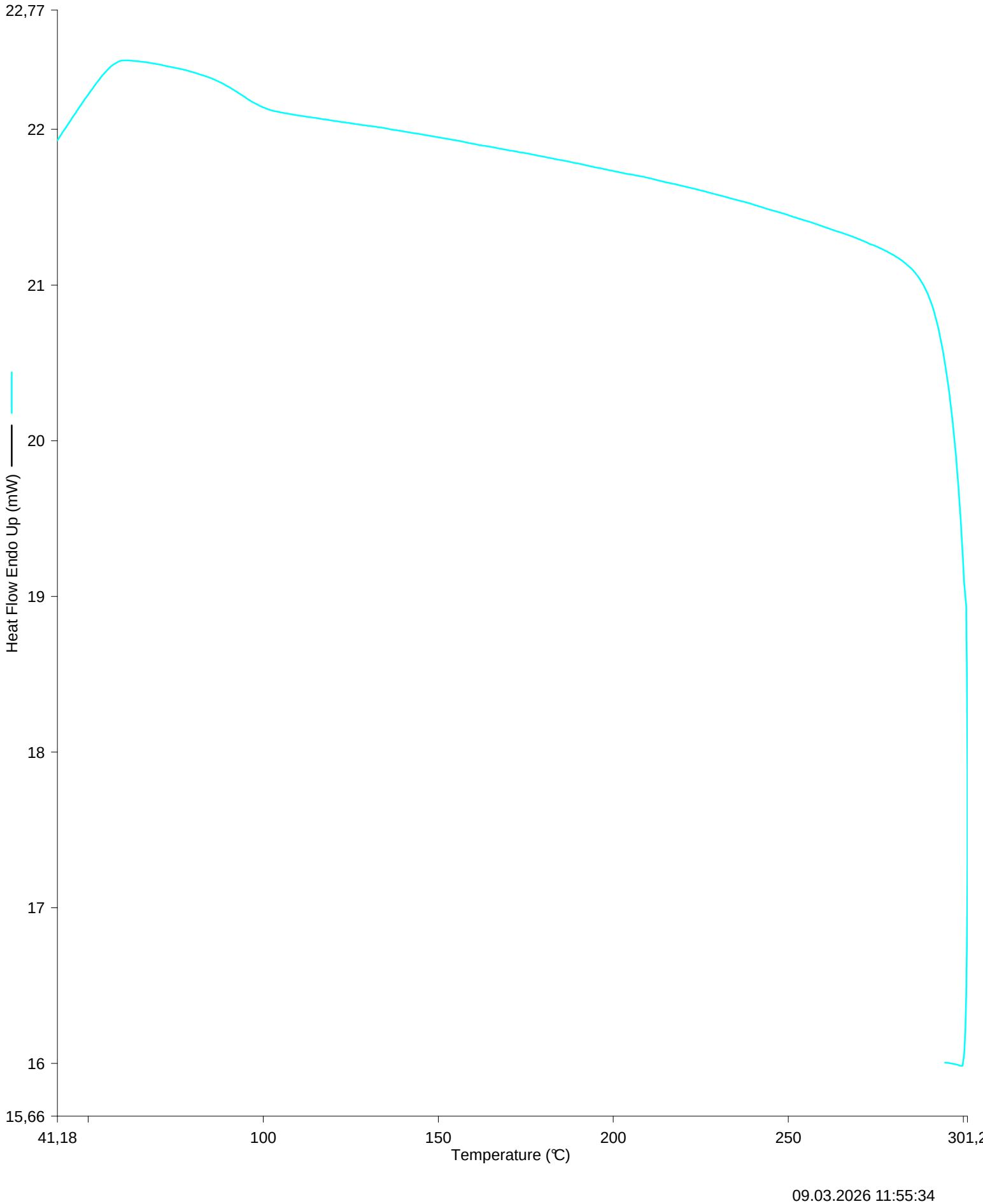
1) Heat from 30.00℃ to 300.00℃ at 20.00℃/min
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

| | |
|---|---|
| Filename: | C:\Users\DSC\Desktop\MD\G7\isops.d6d |
| Operator ID: | MD |
| Sample ID: | G7-iso-PS |
| Sample Weight: | 2.328 mg |
| Comment: | |

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1) Heat from 30.00℃ to 300.00℃ at 20.00℃/min
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

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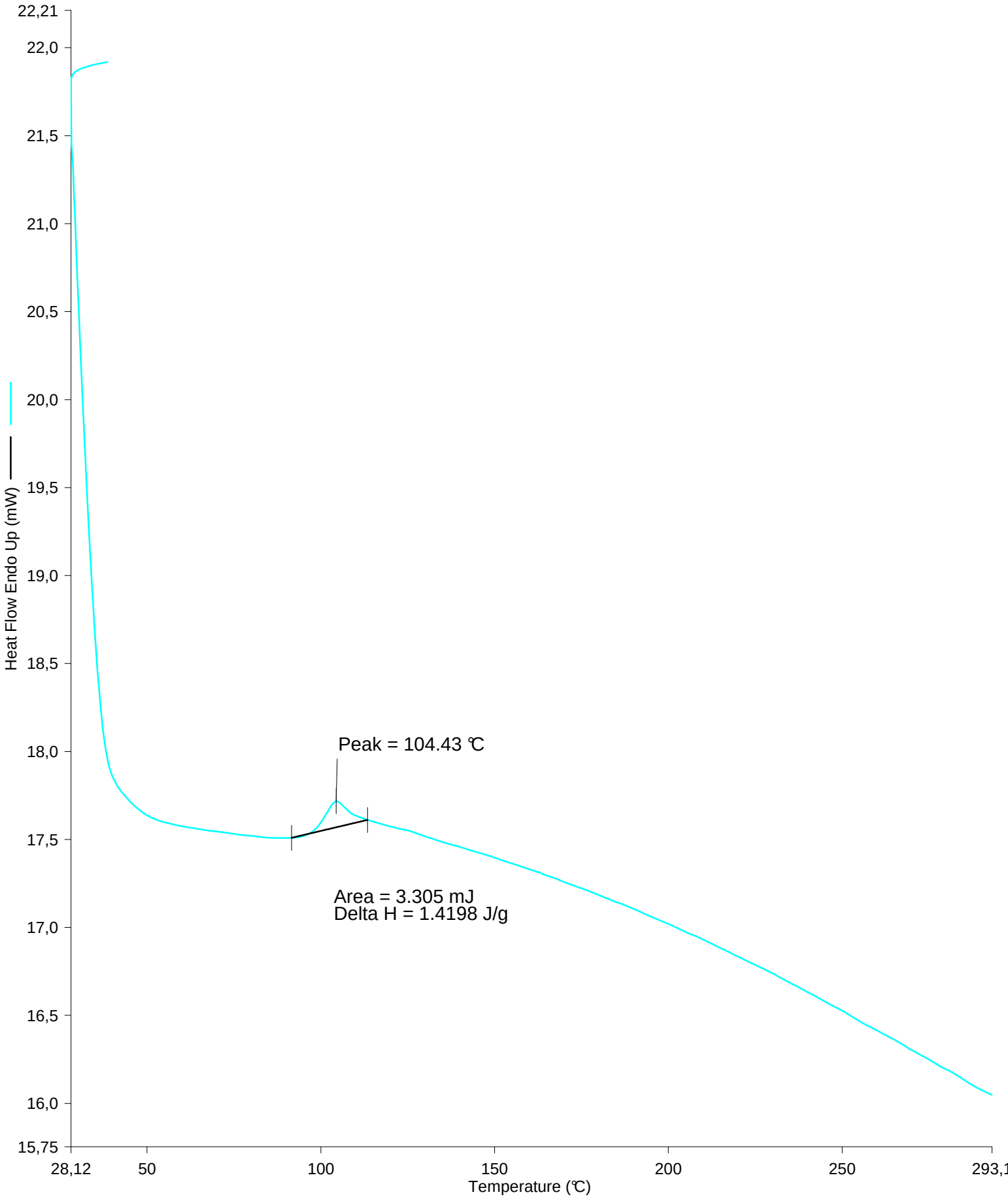
|---|---|
| Filename: | C:\Users\DSC\Desktop\MD\G7\isops.d6d |
| Operator ID: | MD |
| Sample ID: | G7-iso-PS |
| Sample Weight: | 2.328 mg |
| Comment: | |

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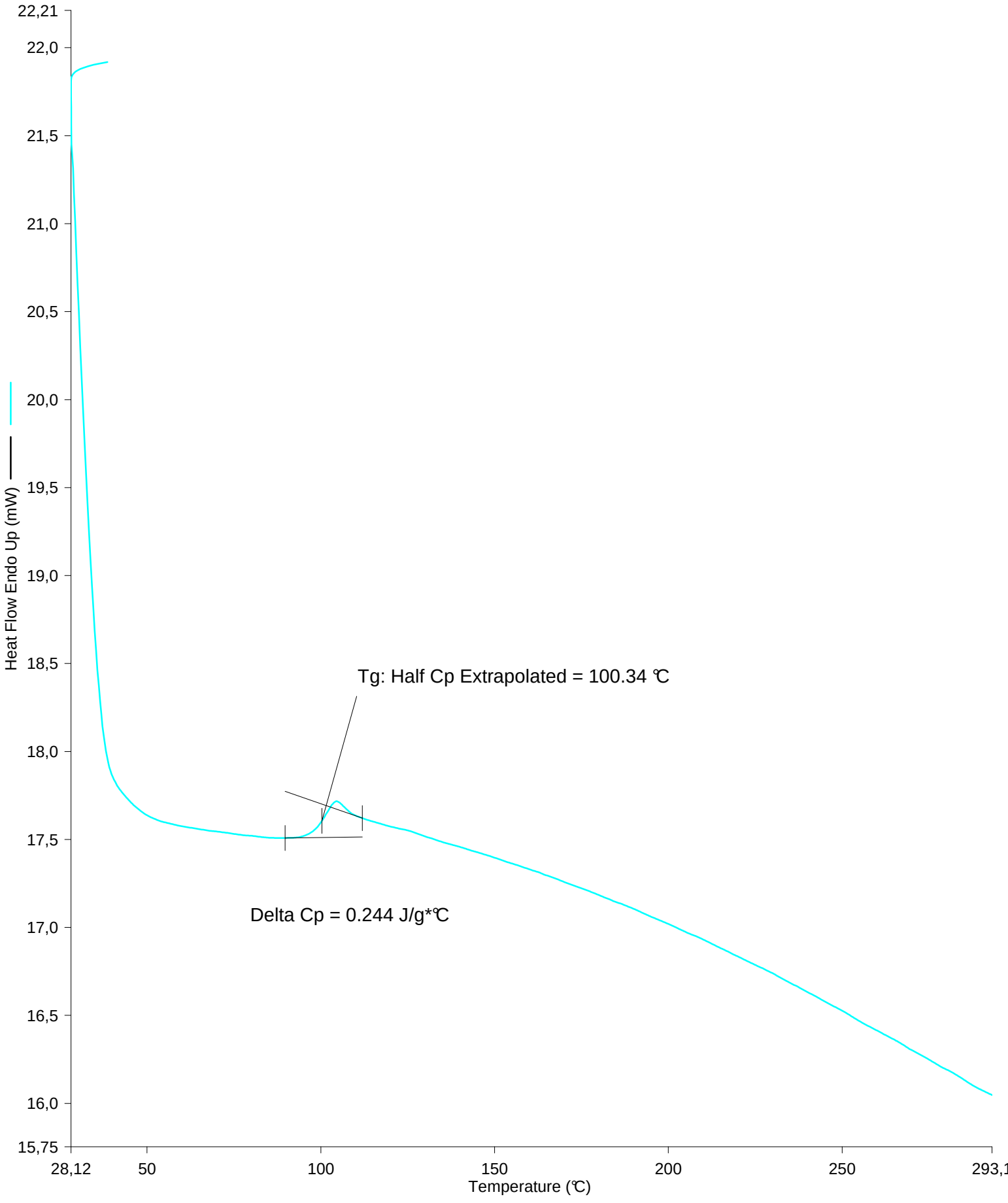
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

| | |
|---|---|
| Filename: | C:\Users\DSC\Desktop\MD\G7\isops.d6d |
| Operator ID: | MD |
| Sample ID: | G7-iso-PS |
| Sample Weight: | 2.328 mg |
| Comment: | |

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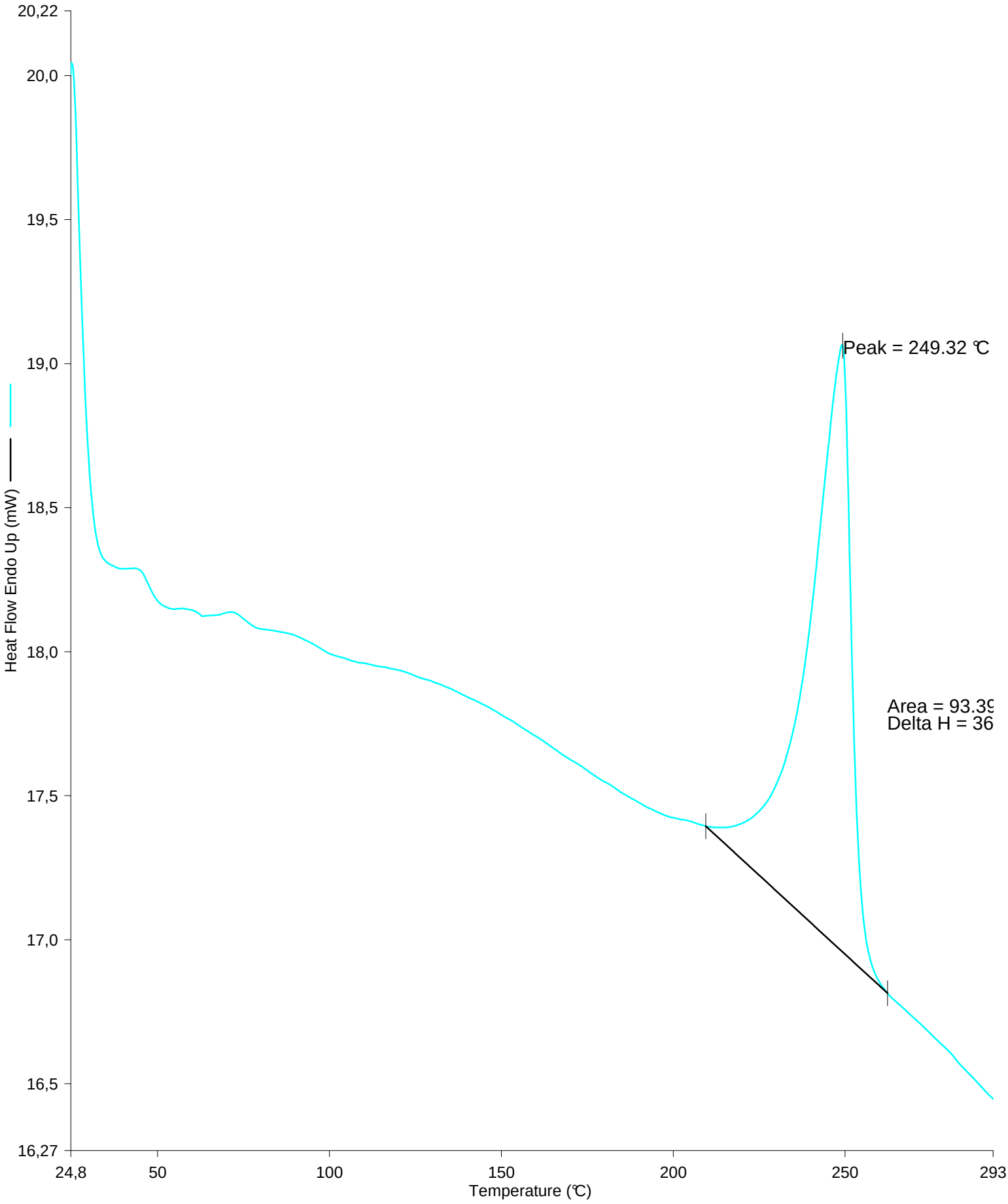
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

| | |
|---|---|
| Filename: | C:\Users\DSC\Desktop\MD\G7\PET.d6d |
| Operator ID: | MD |
| Sample ID: | G7-PET |
| Sample Weight: | 2.530 mg |
| Comment: | |

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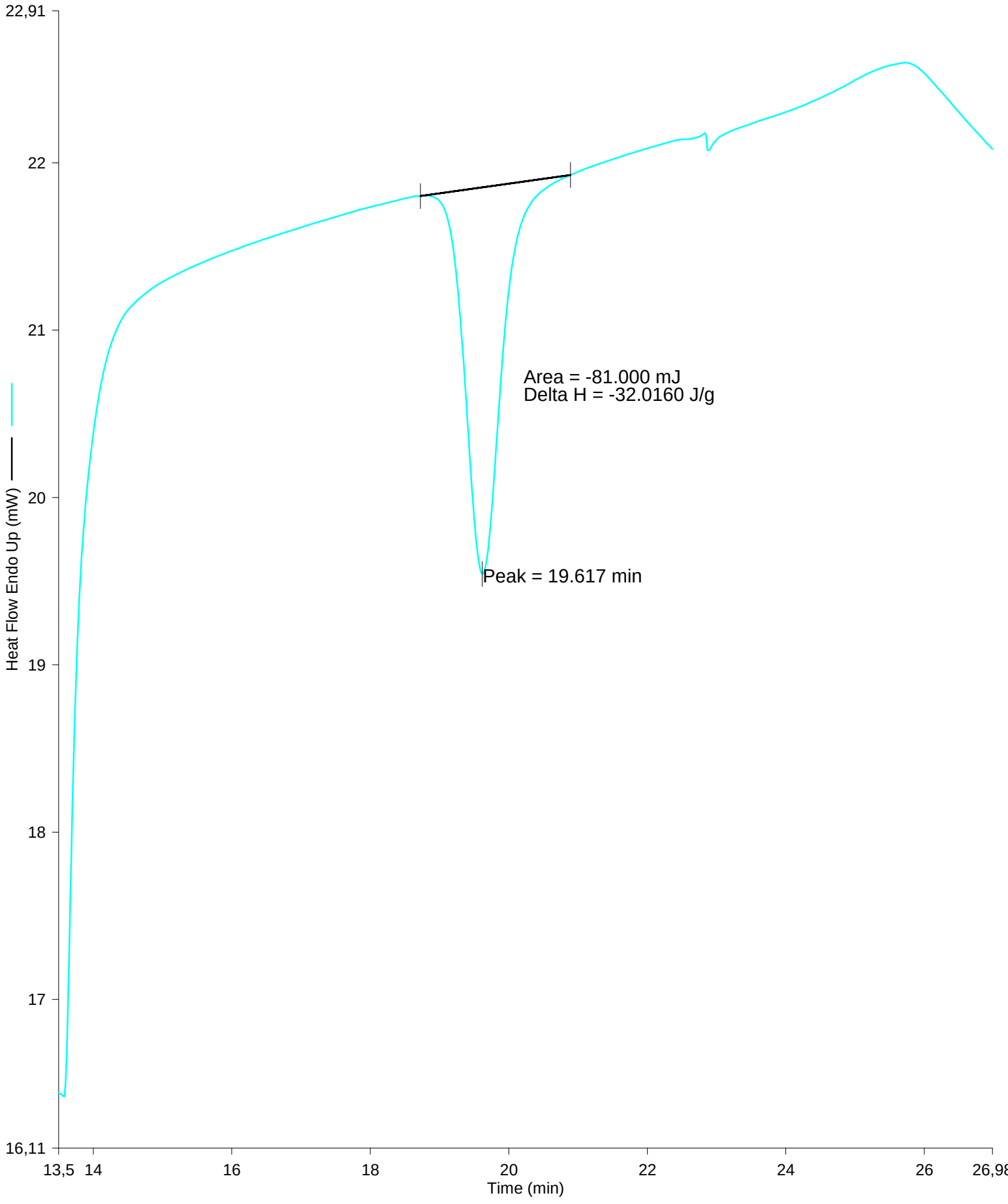
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

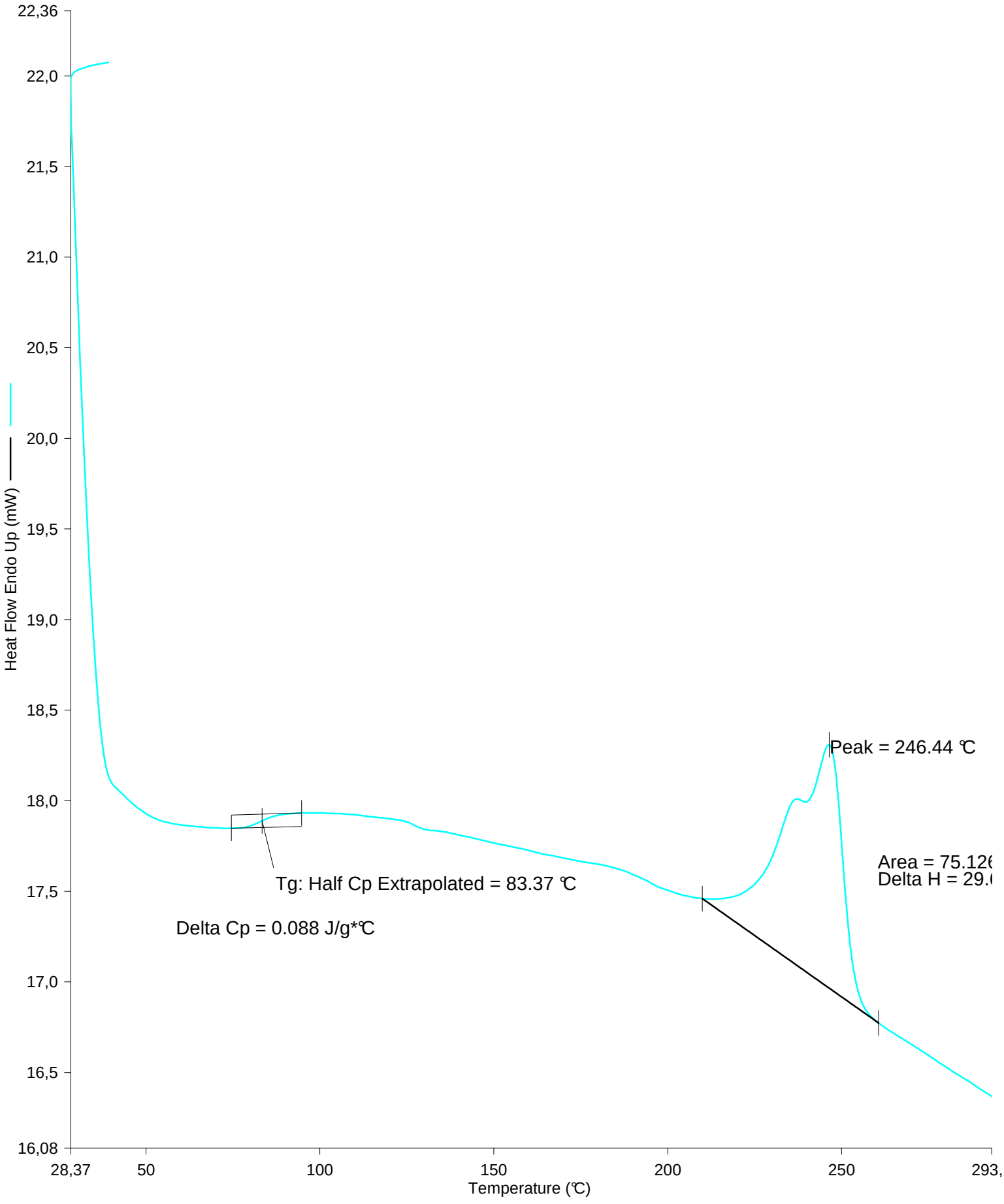
| | |
|---|---|
| Filename: | C:\Users\DSC\Desktop\MD\G7\PET.d6d |
| Operator ID: | MD |
| Sample ID: | G7-PET |
| Sample Weight: | 2.530 mg |
| Comment: | |




09.03.2026 10:54:34

1) Heat from 30.00℃ to 300.00℃ at 20.00℃/min
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

| | |
|---|---|
| Filename: | C:\Users\DSC\Desktop\MD\G7\PET.d6d |
| Operator ID: | MD |
| Sample ID: | G7-PET |
| Sample Weight: | 2.530 mg |
| Comment: | |




09.03.2026 10:59:57

1) Heat from 30.00℃ to 300.00℃ at 20.00℃/min
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min